Lab8 k-means cluster

Source: MLL/Lab8 k-means cluster.ipynb

CELL 1 — CODE

```
import os
os.environ["OMP_NUM_THREADS"] = "1"
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

CELL 2 — CODE

```
data = pd.read_csv("Mall_Customers.csv")
data.columns = data.columns.str.strip()

print("First 5 rows of the dataset:")
print(data.head())
print("\nColumn names:")
print(data.columns.tolist())
```

CELL 3 — CODE

```
required_columns = ["Annual Income (k$)", "Spending Score (1-100)"]

missing_columns = [col for col in required_columns if col not in data.columns]
if missing_columns:
    raise ValueError(
        f"Missing required column(s): {missing_columns}. "
        f"Available columns: {data.columns.tolist()}"
    )

X = data[required_columns].dropna().values

print("\nSelected Features:")
print(X[:5])
```

CELL 4 — CODE

```
wcss = []

for k in range(1, 11):
    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    model.fit(X)
    wcss.append(model.inertia_)
```

CELL 5 — CODE

```
plt.figure(figsize=(7, 5))

plt.plot(
    range(1, 11),
    wcss,
    marker='o'
)

plt.title("Elbow Method - Mall Customers")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.grid(True)
plt.show()
```

CELL 6 — CODE

```
print("Silhouette Scores")

silhouette_scores = []

for k in range(2, 11):
    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    labels = model.fit_predict(X)
    score = silhouette_score(X, labels)
    silhouette_scores.append(score)

    print("K =", k, " Score =", round(score, 3))
```

CELL 7 — CODE

```
plt.figure(figsize=(7, 5))

plt.plot(
    range(2, 11),
    silhouette_scores,
    marker='o'
)

plt.title("Silhouette Score - Mall Customers")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Silhouette Score")
plt.grid(True)
plt.show()
```

CELL 8 — CODE

```
kmeans = KMeans(
    n_clusters=5,
    random_state=42,
    n_init=10
)
```

CELL 9 — CODE

```
clusters = kmeans.fit_predict(X)
```

CELL 10 — CODE

```
data['Cluster'] = clusters

print("\nClustered Dataset:")
print(data.head(10))
```

CELL 11 — CODE

```
print("\nNumber of Customers in Each Cluster:")
print(data['Cluster'].value_counts().sort_index())
```

CELL 12 — CODE

```
plt.figure(figsize=(8, 6))

plt.scatter(
    X[:, 0],
    X[:, 1],
    c=clusters,
    s=50
)

plt.scatter(
    kmeans.cluster_centers_[:, 0],
    kmeans.cluster_centers_[:, 1],
    marker='X',
    s=200,
    label='Centroids'
)

plt.title("K-Means Clustering - Mall Customers")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
```